Vector: A vector is an object that has both a magnitude and a direction. A vector can be denoted as *A*, **A**, $\vec{A}$, $\underline{A}$. Suppose, $\underline{A} = a\hat{\imath} + b\hat{\jmath} + c\hat{k}$, be a vector in 3-D, where $\hat{\imath}, \hat{\jmath}, \hat{k}$ are unit vectors along X, Y and Z-axis. We can also write it as $\vec{A} = (a, b, c)$.

Magnitude of a vector:

If $\vec{A} = a\hat{\imath} + b\hat{\jmath} - c\hat{k}$ be a vector then the magnitude of $\vec{A}$ is denoted by A or $|\vec{A}|$ and is defined as follows:

$$A = |\vec{A}| = \sqrt{a^2 + b^2 + (-c)^2}$$

Example: If $\vec{B} = 2\hat{\imath} + 4\hat{\jmath} - 5\hat{k}$, then the magnitude of $\vec{B}$ is,

$$B = |\vec{B}| = \sqrt{2^2 + 4^2 + (-5)^2} = \sqrt{4 + 16 + 25} = \sqrt{45}$$

Example: If $\vec{A} = 4\hat{\imath} - 3\hat{\jmath} + 2\hat{k}$, $\vec{B} = 5\hat{\imath} - 4\hat{\jmath} + 3\hat{k}$, $\vec{C} = 2\hat{\imath} - 4j + 2\hat{k}$

Then $\vec{A} - 2\vec{B} + \vec{C} = 4\hat{\imath} - 3\hat{\jmath} + 2\hat{k} - 2(5\hat{\imath} + 4\hat{\jmath} - 5\hat{k}) + 2\hat{\imath} - 4j - 2\hat{k}$.

$=4\hat{\imath} - 3\hat{\jmath} + 2\hat{k} - 10\hat{\imath} - 8\hat{\jmath} + 10\hat{k} + 2\hat{\imath} - 4j - 2\hat{k}$.

$=-4\hat{\imath} - 15\hat{\jmath} + 10\hat{k}$

And $|\vec{A} - \overrightarrow{2B} + \vec{C}| = \sqrt{(-4)^2 + (-15)^2 + 10^2} = \sqrt{341}$.

Vector addition, subtraction and scalar multiplication of vectors:

Suppose, $\underline{A} = 4\hat{\imath} - 3\hat{\jmath} + 2\hat{k}$, $\underline{B} = 5\hat{\imath} + 4\hat{\jmath} - 5\hat{k}$ and $\underline{C} = 2\hat{\imath} - 4j - 2\hat{k}$ are three vectors. Find the magnitude of $\underline{A} + \underline{B} + \underline{C}$ *and* $(3\underline{B} - 5\underline{C})$.

Then, $\underline{A} + \underline{B} + \underline{C} = 4\hat{\imath} - 3\hat{\jmath} + 2\hat{k} + 5\hat{\imath} + 4\hat{\jmath} - 5\hat{k} + 2\hat{\imath} - 4j - 2\hat{k} = 11\hat{\imath} - 3\hat{\jmath} - 5\hat{k}$

$|\underline{A} + \underline{B} + \underline{C}| = \sqrt{(11)^2 + (-3)^2 + (-5)^2} = \sqrt{155}$

$3\underline{B} - 5\underline{C} = 3.(5\hat{\imath} + 4\hat{\jmath} - 5\hat{k}) - 5.(2\hat{\imath} - 4j - 2\hat{k}) = 15\hat{\imath} + 12\hat{\jmath} - 15\hat{k} - 10\hat{\imath} + 20\hat{\jmath} + 10\hat{k}$

$= 5\hat{\imath} + 32\hat{\jmath} - 5\hat{k}$

The magnitude of $(3\underline{B} - 5\underline{C})$ is,

$$|3\underline{B} - 5\underline{C}| = \sqrt{5^2 + 32^2 + (-5)^2} = \sqrt{25 + 1024 + 25} = \sqrt{1074}$$

Vector product: There are two types of vector product. (i) Dot product,(ii) Cross product

Dot product: If $\underline{A} = a\hat{\imath} + b\hat{\jmath} + c\hat{k}$ and $\underline{B} = h\hat{\imath} + l\hat{\jmath} + m\hat{k}$ are two vectors then the dot product of $\underline{A}\ and\ \underline{B}\ is\ denoted\ by\ \underline{A}.\underline{B}$ and is defined as follows:

$\underline{A}.\underline{B} = (a\hat{\imath} + b\hat{\jmath} + c\hat{k}).(h\hat{\imath} + l\hat{\jmath} + m\hat{k}) = ah + bl + cm$

Since, $\hat{\imath}.\hat{\imath} = \hat{\jmath}.\hat{\jmath} = \hat{k}.\hat{k} = 1\ and\ \hat{\imath}.\hat{\jmath} = \hat{\jmath}.\hat{k} = \hat{k}.\hat{\imath} = 0$

If $\underline{A}\ and\ \underline{B}$ are two vectors and θ be the angle between them then

$\underline{A}.\underline{B} = |\underline{A}||\underline{B}|\ cos\ \theta$ or, $|\underline{A}||\underline{B}|cos\ \theta = \underline{A}.\underline{B}\ \ or,\ \ cos\ \theta = \frac{\underline{A}.\underline{B}}{|\underline{A}||\underline{B}|}$ or, $\theta = cos^{-1}\left(\frac{\underline{A}.\underline{B}}{|\underline{A}||\underline{B}|}\right)$

Example: Suppose, $\underline{A} = 4\hat{\imath} - 3\hat{\jmath} + 2\hat{k},\quad \underline{B} = 5\hat{\imath} + 4\hat{\jmath} - 5\hat{k}\ \ and\ \ \underline{C} = 2\hat{\imath} - 4j - 2\hat{k}$

(i) Find the value of $(\underline{A} + \underline{B}).(\underline{B} - \underline{C})$

(ii) Show that $(\underline{A} - \underline{B}).\underline{C} = \underline{A}.\underline{C} - \underline{B}.\underline{C}$

(iii) Show that $(\underline{A} + \underline{B}).\underline{C} = \underline{A}.\underline{C} + \underline{B}.\underline{C}$

(iv) Find the angle between $\underline{A}\ and\ \underline{B}$ and between $\underline{B}\ and\ \underline{C}$

(v) Show that $(\underline{A} + \underline{B}).(\underline{A} - \underline{B}) = |\underline{A}|^2 - |\underline{B}|^2$

Solution: (vi): $\underline{A} + \underline{B} = 9\hat{\imath} + \hat{\jmath} - 3\hat{k}, (\underline{A} - \underline{B}) = -\hat{\imath} - 7\hat{\jmath} + 7\hat{k}$

L.H.S=$(\underline{A} + \underline{B}).(\underline{A} - \underline{B}) = (9\hat{\imath} + \hat{\jmath} - 3\hat{k})(\cdot - \hat{\imath} - 7\hat{\jmath} + 7\hat{k}) = -9 - 7 - 21 = -37$

Again $|\underline{A}| = \sqrt{29}$ and $|B| = \sqrt{5^2 + 4^2 + (-5)^2} = \sqrt{66}$

R.H.S=$|\underline{A}|^2 - |\underline{B}|^2 = 29 - 66 = -37$

$\underline{A}.\underline{B} = (4\hat{\imath} - 3\hat{\jmath} + 2\hat{k}) = 20 - 12 - 10 = 20 - 22 = -2$

Solution: (i) $\underline{A} + \underline{B} = 4\hat{\imath} - 3\hat{\jmath} + 2\hat{k} + 5\hat{\imath} + 4\hat{\jmath} - 5\hat{k} = 9\hat{\imath} + \hat{\jmath} - 3\hat{k}$

$(\underline{B} - \underline{C}) = 5\hat{\imath} + 4\hat{\jmath} - 5\hat{k} - 2\hat{\imath} + 4j + 2\hat{k} = 3\hat{\imath} + 8j - 3\hat{k}$

So, $(\underline{A} + \underline{B}).(\underline{B} - \underline{C}) = (9\hat{\imath} + \hat{\jmath} - 3\hat{k}).(3\hat{\imath} + 8j - 3\hat{k})$

$= 27 + 8 + 9 = 44$

Solution (iv): Let θ be the angle between $\underline{A}\ and\ \underline{B}$.

We know that, $\underline{A}.\underline{B} = |\underline{A}||\underline{B}|\ cos\ cos\ \theta$

r, $\theta = cos^{-1}\left(\frac{\underline{A}.\underline{B}}{|\underline{A}||\underline{B}|}\right)$……(i)

Now, $\underline{A}.\underline{B} = -2$, $|\underline{A}| = \sqrt{29}$ & $|\underline{B}| = \sqrt{66}$

From (i) we get, $\theta = cos^{-1}\left(\frac{-2}{\sqrt{29}.\sqrt{66}}\right)$

Cross product: If $\underline{A} = a\hat{\imath} + b\hat{\jmath} + c\hat{k}$ and $\underline{B} = h\hat{\imath} + l\hat{\jmath} + m\hat{k}$ are two vectors then the cross product of $\underline{A}\ and\ \underline{B}\ is\ denoted\ by\ \underline{A} \times \underline{B}$ and is defined as follows:

$$\underline{A} \times \underline{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ a & b & c \\ h & l & m \end{vmatrix} = (bm - cl)\hat{\imath} - (am - ch)\hat{\jmath} + (al - bh)\hat{k}$$

Example: If $\underline{A} = 4\hat{\imath} - 3\hat{\jmath} + 2\hat{k}$ and $\underline{B} = 5\hat{\imath} + 4\hat{\jmath} - 5\hat{k}$ are two vectors then find $\left(\underline{A} \times \underline{B}\right)$.

$$(\underline{A} \times \underline{B}) = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 4 & -3 & 2 \\ 5 & 4 & -5 \end{vmatrix} = 7\hat{\imath} + 30\hat{\jmath} + 31\hat{k}$$

$$\left(\underline{B} \times \underline{A}\right) = -7\hat{\imath} - 30\hat{\jmath} - 31\hat{k} = -\left(7\hat{\imath} + 30\hat{\jmath} + 31\hat{k}\right) = -(\underline{A} \times \underline{B})$$

Again, If $\underline{A}\ and\ \underline{B}$ are two vectors and θ be the angle between them then

$$\underline{A} \times \underline{B} = |\underline{A}||\underline{B}| sin\ \theta$$

Unit vector: If $\underline{A}$ be any vector and $|\underline{A}|$ be the magnitude of $\underline{A}$ then the unit vector along $\underline{A}$ is denoted by $\hat{a}$ and is defined as follows $\hat{a} = \frac{\underline{A}}{|\underline{A}|}$; $|\hat{a}| = 1$.

The unit vector along $(\underline{A} \times \underline{B})$ is = $\frac{(\underline{A} \times \underline{B})}{|\underline{A} \times \underline{B}|}$

Question: If $\underline{A} = 4\hat{\imath} - 3\hat{\jmath} + 2\hat{k}\ and\ \underline{B} = 5\hat{\imath} - 4\hat{\jmath} - 5\hat{k}$ then find the unit vector along $(\underline{A} + \underline{B})$.

Solution: Given, $\underline{A} = 4\hat{\imath} - 3\hat{\jmath} + 2\hat{k}$ and $\underline{B} = 5\hat{\imath} + 4\hat{\jmath} - 5\hat{k}$

Now $\underline{A} + \underline{B} = 4\hat{\imath} - 3\hat{\jmath} + 2\hat{k} + 5\hat{\imath} + 4\hat{\jmath} - 5\hat{k} = 9\hat{\imath} + \hat{\jmath} - 3\hat{k}$

And $|\underline{A} + \underline{B}| = \sqrt{9^2 + 1^2 + (-3)^2} = \sqrt{81 + 1 + 9} = \sqrt{91}$

So, the unit vector along $(\underline{A} + \underline{B})$ is = $\frac{(\underline{A} + \underline{B})}{|\underline{A} + \underline{B}|} = \frac{9\hat{\imath} + \hat{\jmath} - 3\hat{k}}{\sqrt{91}} = \frac{9}{\sqrt{91}}\hat{\imath} + \frac{1}{\sqrt{91}}\hat{\jmath} - \frac{3}{\sqrt{91}}\hat{k}$

Question: If $\underline{P} = 4\hat{\imath} - 2\hat{\jmath} + 3\hat{k}$, $\underline{Q} = 7\hat{\imath} + 2\hat{\jmath} - 5\hat{k}\ and\ \underline{R} = 3\hat{\imath} - 4j - \hat{k}$.

Then, find the unit vector along $\left(\underline{P} \times \underline{Q}\right)$, $\left(\underline{P} + \underline{Q} - \underline{R}\right)$, $(\underline{P} - 2\underline{Q} - 3\underline{R})$ and $(\underline{P} + \underline{Q} + \underline{R})$

Theorem: If $\underline{A}$ and $\underline{B}$ are two vectors and θ be the angle between them then show that $|\underline{A}.\underline{B}|^2 + |\underline{A} \times \underline{B}|^2 = |\underline{A}|^2.|\underline{B}|^2$

Proof: We know that, If $\underline{A}$ *and* $\underline{B}$ are two vectors and θ be the angle between them then

$$\underline{A}.\underline{B} = |\underline{A}||\underline{B}|cos\,\theta \quad \text{……..(i)}$$

$$\& \quad \underline{A} \times \underline{B} = |\underline{A}||\underline{B}|sin\,\theta \quad \text{……..(ii)}$$

Squaring and Adding (i) & (ii) we get,

$$(\underline{A}.\underline{B})^2 + (\underline{A} \times \underline{B})^2 = |\underline{A}|^2.|\underline{B}|^2 sin^2\theta + |\underline{A}|^2.|\underline{B}|^2 cos^2\theta$$

$$= |\underline{A}|^2.|\underline{B}|^2(sin^2\theta + cos^2\theta)$$

Or $(\underline{A}.\underline{B})^2 + (\underline{A} \times \underline{B})^2 = |\underline{A}|^2.|\underline{B}|^2$ [Since $sin^2\theta + cos^2\theta = 1$]

$\therefore \quad |\underline{A}.\underline{B}|^2 + |\underline{A} \times \underline{B}|^2 = |\underline{A}|^2.|\underline{B}|^2$

Since in vector analysis $(\underline{A}.\underline{B})^2 = |\underline{A}.\underline{B}|^2$ & $(\underline{A} \times \underline{B})^2 = |\underline{A} \times \underline{B}|^2$.

So, $|\underline{A}.\underline{B}|^2 + |\underline{A} \times \underline{B}|^2 = |\underline{A}|^2.|\underline{B}|^2$ **(Showed)**

Question: If $\underline{A} = \hat{\imath} + 3\hat{\jmath} + \hat{k}$, $\underline{B} = 2\hat{\imath} + 4\hat{\jmath} - 5\hat{k}$, prove that $|\underline{A}.\underline{B}|^2 + |\underline{A} \times \underline{B}|^2 = |\underline{A}|^2.|\underline{B}|^2$

Solution: Given, $\underline{A} = \hat{\imath} + 3\hat{\jmath} + \hat{k}$, $\underline{B} = 2\hat{\imath} + 4\hat{\jmath} - 5\hat{k}$,

So, $|\underline{A}| = \sqrt{1^2 + 3^2 + 1^2} = \sqrt{1 + 9 + 1} = \sqrt{11}$

And $|\underline{B}| = \sqrt{2^2 + 4^2 + (-5)^2} = \sqrt{4 + 16 + 25} = \sqrt{45}$.

Also, $\underline{A}.\underline{B} = (\hat{\imath} + 3\hat{\jmath} + \hat{k}).(2\hat{\imath} + 4\hat{\jmath} - 5\hat{k}) = 2 + 12 - 5 = 9$

And $\underline{A} \times \underline{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 3 & 1 \\ 2 & 4 & -5 \end{vmatrix} = (-15 - 4)\hat{\imath} — 5 - 2\hat{\jmath} + (4 - 6)\hat{k} = -19\hat{\imath} + 7\hat{\jmath} - 2\hat{k}$

$|\underline{A} \times \underline{B}| = \sqrt{(-19)^2 + 7^2 + (-2)^2} = \sqrt{414}$

L.H.S=$|\underline{A}.\underline{B}|^2 + |\underline{A} \times \underline{B}|^2 = 81 + 414 = 495$ and R.H.S= $|\underline{A}|^2.|\underline{B}|^2$ = 11.45=495

Hence, $|\underline{A}.\underline{B}|^2 + |\underline{A} \times \underline{B}|^2 = |\underline{A}|^2.|\underline{B}|^2$ **(Proved)**

Exercise: If $\underline{A} = 2\hat{\imath} - \hat{\jmath} + 4\hat{k}$, $\underline{B} = \hat{\imath} - 4\hat{\jmath} + 3\hat{k}$, prove that $|\underline{A}.\underline{B}|^2 + |\underline{A} \times \underline{B}|^2 = |\underline{A}|^2.|\underline{B}|^2$

Vector differentiation:

Del/delta: A vector valued function is denoted and defined as follows:

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}$$

Gradient: If φ be a scalar valued function then the gradient of φ is denoted by grad φ or $\underline{\nabla}\varphi$ and is defined as follows:

$$\text{grad } \varphi = \vec{\nabla}\varphi = \left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right)\varphi = \frac{\partial \varphi}{\partial x}\hat{\imath} + \frac{\partial \varphi}{\partial y}\hat{\jmath} + \frac{\partial \varphi}{\partial z}\hat{k}$$

Question: Suppose $\varphi = xyz$ then find the Gradient of φ at the point (2, 0, 1).

Solution: Given, $\varphi = xyz$

So, **grad** $\varphi = \vec{\nabla}\varphi = \left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right) = \frac{\partial \varphi}{\partial x}\hat{\imath} + \frac{\partial \varphi}{\partial y}\hat{\jmath} + \frac{\partial \varphi}{\partial z}\hat{k}$

$= \frac{\partial}{\partial x} xyz\,\hat{\imath} + \frac{\partial}{\partial y} xyz\,\hat{\jmath} + \frac{\partial}{\partial z}(xyz)\,\hat{k}$

$= yz.1\,\hat{\imath} + xz.1\,\hat{\jmath} + xy.1\,\hat{k}$

$\vec{\nabla}\varphi = yz\,\hat{\imath} + xz\,\hat{\jmath} + xy\,\hat{k}$

At the point (2, 0, 1), grad $\varphi = \vec{\nabla}\varphi = 0 \cdot 1\,\hat{\imath} + 2 \cdot 1\,\hat{\jmath} + 2 \cdot 0\,\hat{k} = 2\,\hat{\jmath}$

Question: Suppose $\varphi = (xz - y^2)$ then find the Gradient of φ at the point (3, -1, 1).

Solution: $\boldsymbol{grad}\ \varphi = \vec{\nabla}\varphi = \frac{\partial}{\partial x}(xz - y^2)\hat{\imath} + \frac{\partial}{\partial y}(xz - y^2)\hat{\jmath} + \frac{\partial}{\partial z}(xz - y^2)\hat{k}$

$= (z.1 - 0)\hat{\imath} + (0 - 2y)\hat{\jmath} + (x.1 - 0)\hat{k}$

$= z\hat{\imath} - 2y\hat{\jmath} + x\hat{k}$

At the point (3, -1, 1), grad $\varphi = \underline{\nabla}\varphi = 1\,\hat{\imath} - 2(-1)\hat{\jmath} + 3\,\hat{k} = \hat{\imath} + 2\,\hat{\jmath} + 3\,\hat{k}$

Question: Suppose $\varphi = 2xz - yz$ then find the Gradient of φ at the point (2, -1, 0).

$grad\ \varphi = \vec{\nabla}\varphi = \frac{\partial}{\partial x}(2xz - yz)\,\hat{\imath} + \frac{\partial}{\partial y}(2xz - yz)\,\hat{\jmath} + \frac{\partial}{\partial z}(2xz - yz)\,\hat{k}$

$= (2z.1 - 0)\hat{\imath} + (0 - z.1)\hat{\jmath} + (2x.1 - y.1)\hat{k}$

$= 2z\,\hat{\imath} - z\,\hat{\jmath} + (2x - y)\hat{k}$

At the point (2, -1, 0), grad $\varphi = \vec{\nabla}\varphi = 2 \cdot 0\,\hat{\imath} - 0\,\hat{\jmath} + (2 \cdot 2 + 1)\hat{k} = 5\,\hat{k}$

Divergence: If $\vec{A} = f\hat{\imath} + g\hat{\jmath} + h\hat{k}$ be a vector valued function then the divergence of $\vec{A}$ is denoted by div $\vec{A}$ or $\vec{\nabla} \cdot \vec{A}$ and is defined as follows:

$$\text{div } \underline{A} \text{ or } \vec{\nabla} \cdot \vec{A} = \left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right) \cdot \left(f\hat{\imath} + g\hat{\jmath} + h\hat{k}\right) = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}$$

Question-1: If $\vec{A} = xz\,\hat{\imath} - xy\,\hat{\jmath} + yz\,\hat{k}$ then find the Divergence of $\vec{A}$ at the point (1,-1, 2).

Solution: Given, $\vec{A} = xz\,\hat{\imath} - xy\,\hat{\jmath} + yz\,\hat{k}$

Now, div $\vec{A}$ or $\vec{\nabla} \cdot \vec{A} = \left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right) \cdot (xz\,\hat{\imath} - xy\,\hat{\jmath} + yz\,\hat{k})$

$$= \frac{\partial}{\partial x}(xz) - \frac{\partial}{\partial y}(xy) + \frac{\partial}{\partial z}(yz) = z - x + y$$

At the point (1,-1, 2), div $\underline{A}$ or $\vec{\nabla} \cdot \vec{A} = 2 - 1 - 1 = 0$

Question-2: If $\vec{A} = x^2\hat{\imath} - z^2\hat{\jmath} - y^2z\hat{k}$ then find the Divergence of $\underline{A}$ at the point (-3,0, 2).

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial y}(z^2) - \frac{\partial}{\partial z}(y^2z) = 2x - 0 - y^2 = 2x - y^2$$

At the point (-3, 0, 2) ,div $\vec{A}$ or $\vec{\nabla} \cdot \vec{A} = 2(-3) - 0 = -6$

Exercise: Let $\vec{P} = x^2yz\,\hat{\imath} - (xz - y^2)\hat{\jmath} - yz\,\hat{k}$, find the Divergence of $\vec{P}$ at the point (2, 1, -3).

Curl: If $\vec{A} = f\hat{\imath} + g\hat{\jmath} + h\hat{k}$ be a vector valued function then the curl of $\vec{A}$ is denoted by curl $\vec{A}$ or $\vec{\nabla} \times \vec{A}$ and is defined as follows:

$$\text{curl } \vec{A} = \vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & h \end{vmatrix} = \hat{\imath}\left(\frac{\partial h}{\partial y} - \frac{\partial g}{\partial z}\right) - \left(\frac{\partial h}{\partial x} - \frac{\partial f}{\partial z}\right)\hat{\jmath} + \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right)\hat{k}$$

Question: If $\vec{A} = x^2\hat{\imath} - z^2j + y^2\hat{k}$ then find the Curl of $\vec{A}$ at the point (1,0, 1).

Solution: Given, $\vec{A} = x^2\hat{\imath} - z^2j + y^2\hat{k}$.

$$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & -z^2 & y^2 \end{vmatrix}$$

$= \left(\frac{\partial}{\partial y}y^2 + \frac{\partial}{\partial z}z^2\right)\hat{\imath} - \left(\frac{\partial}{\partial x}y^2 - \frac{\partial}{\partial z}x^2\right)\hat{\jmath} + \left(-\frac{\partial}{\partial x}z^2 - \frac{\partial}{\partial y}x^2\right)\hat{k}$

$= (2y + 2z)\hat{\imath} - (0 - 0)\hat{\jmath} + (0 - 0)\hat{k}$

$= (2y + 2z)\hat{\imath}$

At the point (1,0, 1), $\vec{A}$ or $\vec{\nabla} \times \vec{A} = (2.0 + 2.1)\hat{\imath} = 2\,\hat{\imath}$

Exercise: If $\vec{A} = xy\hat{\imath} - yz\hat{\jmath} + z^2\hat{k}$ then find the Curl of $\underline{A}$ at the point (-1,1, 2).

Question: Suppose, $\vec{P} = xy\,\hat{\imath} - yz\,\hat{\jmath} + z^2\,\hat{k}$, $\vec{Q} = y\hat{\imath} + 2x\hat{\jmath} - 3z\hat{k}$, $\vec{R} = x^2\hat{\imath} - z^2\hat{\jmath} - y^2\hat{k}$ and $\varphi = xyz$. Show that $(i)\ (\vec{P}\cdot\vec{\nabla})\varphi = (\vec{\nabla}\varphi)\cdot\vec{P}$ and $(ii)(\vec{R}\cdot\vec{\nabla})\varphi = (\vec{\nabla}\varphi)\cdot\vec{R}$ (***yourself***)

Solution: (i)

Here, $\vec{P}\cdot\vec{\nabla} = \left(xy\hat{\imath} - yz\hat{\jmath} + z^2\hat{k}\right).\left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right) = xy\frac{\partial}{\partial x} - yz\frac{\partial}{\partial y} + z^2\frac{\partial}{\partial z}$

Therefore $(\vec{P}\cdot\vec{\nabla})\varphi = \left(xy\frac{\partial}{\partial x} - yz\frac{\partial}{\partial y} + z^2\frac{\partial}{\partial z}\right)(xyz)$

$= xy\frac{\partial}{\partial x}(xyz) - yz\frac{\partial}{\partial y}(xyz) + z^2\frac{\partial}{\partial z}(xyz)$

$= xy.yz - yz.xz + z^2.xy = xy^2z$

Again, $\vec{\nabla}\varphi = \left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right)(xyz)$

$= \frac{\partial}{\partial x}(xyz)\,\hat{\imath} + \frac{\partial}{\partial y}(xyz)\,\hat{\jmath} + \frac{\partial}{\partial z}(xyz)\hat{k}$

$= yz\,\hat{\imath} + xz\,\hat{\jmath} + xy\,\hat{k}$

$(\vec{\nabla}\varphi)\cdot\vec{P} = \left(yz\,\hat{\imath} + xz\,\hat{\jmath} + xy\,\hat{k}\right).(xy\hat{\imath} - yz\hat{\jmath} + z^2\hat{k})$

$= xy^2z - xyz^2 + xyz^2 = xy^2z.$

Thus $(\vec{P}\cdot\vec{\nabla})\varphi = (\vec{\nabla}\varphi)\cdot\vec{P}$ **(Showed)**

Question: Prove $(i)\ \ \vec{\nabla}\cdot(\vec{Q} + \vec{R}) = \vec{\nabla}\cdot\ \vec{Q} + \vec{\nabla}\cdot\vec{R}$ $(ii)\vec{\nabla}\cdot(\vec{P} + \vec{Q}) = \vec{\nabla}\cdot\vec{P} + \vec{\nabla}\cdot\vec{Q}$ (***youself***)

Solution: Here $\vec{Q} + \vec{R} = y\hat{\imath} + 2x\hat{\jmath} - 3z\hat{k} + x^2\hat{\imath} - z^2j - y^2\hat{k}$

$= (x^2 + y)\,\hat{\imath} + (2x - z^2)\,\hat{\jmath} - (\,3z + y^2)\,\hat{k}$

$\vec{\nabla} \cdot \left(\vec{Q} + \vec{R}\right) = \left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right).\{(x^2 + y)\,\hat{\imath} + (2x - z^2)\,\hat{\jmath} - (\,3z + y^2)\,\hat{k}\}$

$= \frac{\partial}{\partial x}(x^2 + y) + \frac{\partial}{\partial y}(2x - z^2) - \frac{\partial}{\partial z}(\,3z + y^2) = 2x + 0 - 3 = \;2x \;-\; 3$

Again, $\vec{\nabla} \cdot \;\vec{Q} = (\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}).(\,y\hat{\imath} + 2x\hat{\jmath} - 3z\hat{k}$

$$= \frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(2x) - \frac{\partial}{\partial z}(3z) = 0 - 0 - 3 = -3$$

And $\vec{\nabla} \cdot \;\vec{R} = (\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}).(\,x^2\hat{\imath} \;- z^2 j - y^2\hat{k})$

$$= \frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial y}(z^2) - \frac{\partial}{\partial z}(y^2) = 2x - 0 - 0 = 2x$$

$\vec{\nabla} \cdot \;\vec{Q} + \vec{\nabla} \cdot \vec{R} \;= -3 + 2x = 2x - 3.$

Thus $\;\vec{\nabla} \cdot \left(\vec{Q} + \vec{R}\right) = \vec{\nabla} \cdot \;\vec{Q} + \vec{\nabla} \cdot \vec{R}\;$ **(Proved)**

Question: (a) If $\varphi = x^2yz$ then prove that $\;\vec{\nabla} \times \left(\vec{\nabla}\varphi\right) = 0.$

(b) If $\varphi = xyz$ and $\;\underline{A} = xz\hat{\imath} - y^2\hat{\jmath} + z^2\hat{k}\;$ then show that $\;\vec{\nabla} \cdot \left(\,\vec{\nabla} \times \vec{\mathrm{A}}\right) = 0.$

Solution: (a)

$$\vec{\nabla}\varphi = \left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right)\varphi = \frac{\partial\varphi}{\partial x}\hat{\imath} + \frac{\partial\varphi}{\partial y}\hat{\jmath} + \frac{\partial\varphi}{\partial z}\hat{k}$$

$$= \frac{\partial}{\partial x}(x^2yz)\;\hat{\imath} + \frac{\partial}{\partial y}(x^2yz)\;\hat{\jmath} + \frac{\partial}{\partial z}(x^2yz)\;\hat{k}$$

$$= 2xyz\;\hat{\imath} + x^2z\;\hat{\jmath} + x^2y\,\hat{k}$$

Now, $\vec{\nabla} \times \left(\vec{\nabla}\varphi\right) = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xyz & x^2z & x^2y \end{vmatrix}$

$$= \hat{\imath}\left\{\frac{\partial}{\partial y}(x^2y) - \frac{\partial}{\partial z}(x^2z)\right\} - \hat{\jmath}\left\{\frac{\partial}{\partial x}(x^2y) - \frac{\partial}{\partial z}(2xyz)\right\} + \hat{k}\left\{\frac{\partial}{\partial x}(x^2z) - \frac{\partial}{\partial y}(2xyz)\right\}$$

$$= (x^2 - x^2)\hat{\imath} - (2xy - 2xy)\hat{\jmath} + (2xz - 2xz)\hat{k} = 0.$$

(b) $\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & -y^2 & z^2 \end{vmatrix}$

$$= \hat{\imath}\left(\frac{\partial}{\partial y}z^2 + \frac{\partial}{\partial z}y^2\right) - \hat{\jmath}\left(\frac{\partial}{\partial x}z^2 - \frac{\partial}{\partial z}xz\right) + \hat{k}\left(-\frac{\partial}{\partial x}y^2 - \frac{\partial}{\partial y}xz\right)$$

$$= (0+0)\hat{\imath} - (0 - x.1)\hat{\jmath} + (-0-0)\hat{k} = x\hat{\jmath}$$

Now, $\vec{\nabla} \cdot \left(\vec{\nabla} \times \vec{A}\right) = \left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right).(x\hat{\jmath}) = \frac{\partial}{\partial y}(x) = 0.$

Question: If $\varphi = xyz^2$ and $\vec{A} = x^2\hat{\imath} - z^2\hat{\jmath} - y^2\hat{k}$ then show $\vec{\nabla} \cdot (\varphi\vec{A}) = (\vec{\nabla}\varphi) \cdot \vec{A} + \varphi(\vec{\nabla} \cdot \vec{A})$

Solution: Here $\varphi\vec{A} = xyz^2.\left(x^2\hat{\imath} - z^2\hat{\jmath} - y^2\hat{k}\right) = x^3yz^2\hat{\imath} - xyz^4\hat{\jmath} - xy^3z^2\hat{k}$

So, $\vec{\nabla} \cdot (\varphi\vec{A}) = \left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right).(x^3yz^2\hat{\imath} - xyz^4\hat{\jmath} - xy^3z^2\hat{k})$

$$= \frac{\partial}{\partial x}(x^3yz^2) - \frac{\partial}{\partial y}(xyz^4) - \frac{\partial}{\partial z}(xy^3z^2)$$

$$= 3x^2yz^2 - xz^4 - 2xy^3z$$

Also, $\vec{\nabla}\varphi = \left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right).xyz^2 = \frac{\partial}{\partial x}(xyz^2)\,\hat{\imath} + \frac{\partial}{\partial y}(xyz^2)\,\hat{\jmath} + \frac{\partial}{\partial z}(xyz^2)\,\hat{k}$

$$= yz^2\,\hat{\imath} + xz^2\,\hat{\jmath} + 2xyz\hat{k}$$

Therefore $(\vec{\nabla}\varphi) \cdot \vec{A} = \left(yz^2\,\hat{\imath} + xz^2\,\hat{\jmath} + 2xyz\hat{k}\right) \cdot \left(x^2\hat{\imath} - z^2\hat{\jmath} - y^2\hat{k}\right) = x^2yz^2 - xz^4 - 2xy^3z$

Again, $\varphi(\vec{\nabla} \cdot \vec{A}) = xyz^2\left[\left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right) \cdot \left(x^2\hat{\imath} - z^2\hat{\jmath} - y^2\hat{k}\right)\right]$

$$= xyz^2\left\{\frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial y}(z^2) - \frac{\partial}{\partial z}(y^2)\right\} = 2x^2yz^2$$

$$(\vec{\nabla}\varphi) \cdot \vec{A} + \varphi(\vec{\nabla} \cdot \vec{A}) = 2x^2yz^2 + x^2yz^2 - xz^4 - 2xy^3z = 3x^2yz^2 - xz^4 - 2xy^3z$$

Hence, $\vec{\nabla} \cdot (\varphi\vec{A}) = (\vec{\nabla}\varphi) \cdot \vec{A} + \varphi(\vec{\nabla} \cdot \vec{A})$ **(Showed)**

Vector Integration: If $R(u) = r_1(u)\hat{\imath} + r_2(u)\hat{\jmath} + r_3(u)\hat{k}$ then

$$\int R(u)du = \int\{r_1(u)\hat{\imath} + r_2(u)\hat{\jmath} + r_3(u)\hat{k}\}du.$$

$$= \hat{\imath}\int r_1(u)du + \hat{\jmath}\int r_2(u)du + \hat{k}\int r_3(u)du$$

Question: Suppose $R(u) = 3\hat{\imath} + (u^3 + 4u^7)\hat{\jmath} + u\hat{k}$. Find $\int R(u)du$ and $\int_0^2 R(u)du$.

Solution: Given, $R(u) = 3\hat{\imath} + (u^3 + 4u^7)\hat{\jmath} + u\hat{k}$.

So, $\int R(u)du = \int\{3\hat{\imath} + (u^3 + 4u^7)\hat{\jmath} + u\hat{k}\}du$

$$= 3u\hat{\imath} + \left(\frac{u^4}{4} + 4.\frac{u^8}{8}\right)\hat{\jmath} + \frac{u^2}{2}\hat{k} + c = 3u\hat{\imath} + \left(\frac{u^4}{4} + \frac{u^8}{2}\right)\hat{\jmath} + \frac{u^2}{2}\hat{k} + c$$

Again,

$$\int_0^2 R(u)du = \int_0^2\{3\hat{\imath} + (u^3 + 4u^7)\hat{\jmath} + u\hat{k}\}du$$

$$= \left[3u\hat{\imath} + \left(\frac{u^4}{4} + \frac{u^8}{2}\right)\hat{\jmath} + \frac{u^2}{2}\hat{k}\right]_0^2$$

$$= 3(2-0)\hat{\imath} + \left(\frac{2^4}{4} + \frac{2^8}{2} - 0 - 0\right)\hat{\jmath} + (\frac{2^2}{2} - 0)\hat{k} = 6\hat{\imath} + 132\hat{\jmath} + 2\hat{k}$$

Exercise:

1. Suppose $R(u) = (2u^3 - 6u)\,\hat{\imath} - 3u^5\,\hat{\jmath} - 32u\,\hat{k}$. Find $\int R(u)du$ and $\int_0^3 R(u)du$.

   Ans: $36\,\hat{\imath} - 192\,\hat{k}$

2. Suppose $P(u) = 2u\hat{\imath} + (u - 3u^4)\hat{\jmath} - u^2\hat{k}$. Find $\int P(u)du$ and $\int_{-1}^2 P(u)du$.

   Ans: $3\,\hat{\imath} - \frac{183}{10}\hat{\jmath} - 3\,\hat{k}$

**** If the position vector of a point is $\underline{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ then $d\underline{r} = dx\hat{\imath} + dy\hat{\jmath} + dz\hat{k}$.**

Question: If $\vec{F} = 2x\,\hat{\imath} - xy^2\,\hat{\jmath} + yz^3\,\hat{k}$, then find the value of $\int \vec{F}.d\vec{r}$

Solution: Here $\int \vec{F}.d\vec{r} = \int(2x\hat{\imath} - xy^2\hat{\jmath} + yz^3\hat{k}).(dx\hat{\imath} + dy\hat{\jmath} + dz\hat{k})$

$$= \int(2x\,dx - xy^2\,dy + yz^3\,dz) = 2\cdot\frac{x^2}{2} - x\cdot\frac{y^3}{3} + y\cdot\frac{z^4}{4} + c$$

$$= x^2 - \frac{xy^3}{3} + \frac{yz^4}{4} + c \quad \text{(Ans:)}$$

Question: If $\vec{F} = xz\hat{\imath} - y^2\hat{\jmath} + mz^5\hat{k}$ and $f = xy^2z$ find the value of $\int \vec{F}.d\vec{r}$ and $\int f.d\vec{r}$

Green's Theorem: Let R be a simply connected plane region whose boundary is a simple, closed, piecewise smooth curve C oriented counterclockwise. If $f(x,y)$ and $g(x,y)$ are continuous and have continuous first partial derivatives on some open set containing R, then

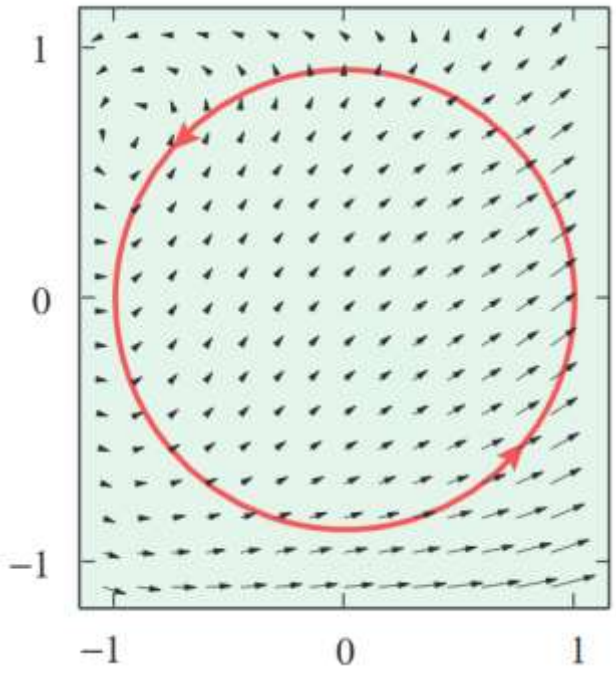
$$\oint_C f(x,y)\,dx + g(x,y)\,dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right) dA$$

Example-1: Find the work done by the force field $\vec{F}(x,y,z) = \left(e^x - y^3\right)\hat{i} + \left(\cos y + x^3\right)\hat{j}$ on a particle that travels once around the unit circle $x^2 + y^2 = 1$ in the counterclockwise direction.

Solution:

The work W performed by the field is

$$\begin{aligned} W &= \oint_C \vec{F}\cdot d\vec{r} \\ &= \oint_C \left(e^x - y^3\right)dx + \left(\cos y + x^3\right)dy \\ &= \iint_R \left[\frac{\partial}{\partial x}\left(\cos y + x^3\right) - \frac{\partial}{\partial y}\left(e^x - y^3\right)\right] dA \\ &= \iint_R \left(3x^2 + 3y^2\right) dA \\ &= 3\iint_R \left(x^2 + y^2\right) dx\,dy \\ &= 3\int_0^{2\pi}\int_0^1 \left(r^2\right) r\,dr\,d\theta \\ &= 3\left[\frac{r^4}{4}\right]_0^1 \left[\theta\right]_0^{2\pi} = \frac{3\pi}{2} \end{aligned}$$



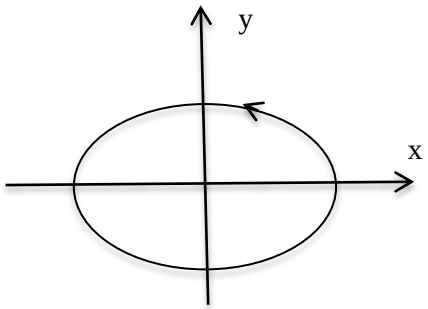
Example-2: Use a line integral to find the area enclosed by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

Solution: The ellipse, with counterclockwise orientation, can be represented parametrically by

$$x=a\cos\theta,\quad y=b\sin\theta,\qquad 0\le\theta\le 2\pi$$

If we denote this curve by C, then the area A enclosed by the ellipse is

$$\begin{aligned}A&=\frac{1}{2}\oint_C x\,dy-y\,dx\\&=\frac{1}{2}\int_0^{2\pi}\left[(-b\sin\theta)(-a\sin\theta)+(a\cos\theta)(b\cos\theta)\right]d\theta\\&=\frac{1}{2}ab\int_0^{2\pi}\left(\sin^2\theta+\cos^2\theta\right)d\theta\\&=\frac{1}{2}ab\int_0^{2\pi}1\,d\theta=\pi ab\end{aligned}$$



Use Green's Theorem to evaluate the integral. In each exercise, assume that the curve C is oriented counterclockwise.

1. $\oint_C \left(x^2-2y^2\right)dx+x\,dy$, where C is the circle $x^2+y^2=9$.

2. $\oint_C y\tan^2 x\,dx+\tan x\,dy$, where C is the circle $x^2+(y+1)^2=1$.

3. $\oint_C \left(x^2-3y\right)dx+3x\,dy$, where C is the circle $x^2+y^2=4$.

4. $\oint_C \left(e^x+y^2\right)dx+\left(e^y+x^2\right)dy$, where C is the boundary of the region between $y=x^2$ & $y=2x$

5. $\oint_C x^2y\,dx-y^2x\,dy$, where C is the boundary of the region in the first quadrant, enclosed between the coordinate axes and the circle $x^2+y^2=16$.

Divergence theorem: Let G be a solid whose surface σ is oriented outward.

If $\vec{F}(x,y,z)=f(x,y,z)\hat{i}+g(x,y,z)\hat{j}+h(x,y,z)\hat{k}$ where f, g, and h have continuous first partial derivatives on some open set containing G, and if n is the outward unit normal on σ, then

$$\iint_\sigma \vec{F}\cdot\hat{n}\,dS=\iiint_G div\,\vec{F}\,dV$$

Example: Use the Divergence Theorem to find the outward flux of the vector field $\vec{F}(x,y,z) = x^3\hat{i} + y^3\hat{j} + z^2\hat{k}$ across the surface of the region that is enclosed by the circular cylinder $x^2 + y^2 = 9$ and the planes $z = 0$ and $z = 2$

Solution: Let σ denote the outward-oriented surface and G the region that it encloses. The divergence of the vector field is $div\,\vec{F} = \frac{\partial}{\partial x}(x^3) + \frac{\partial}{\partial y}(y^3) + \frac{\partial}{\partial z}(z^2) = 3x^2 + 3y^2 + 2z$

Therefore, the flux across σ is

$$\iint_{\sigma} \vec{F}\cdot\hat{n}\,dS = \iiint_{G} (3x^2 + 3y^2 + 2z)\,dV$$

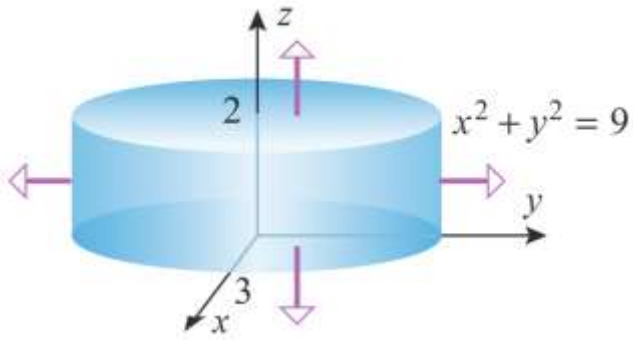
$$= \int_0^{2\pi}\int_0^3\int_0^2 (3r^2 + 2z)\,r\,dz\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^3 \left[3r^3 z + z^2 r\right]_{z=0}^{2} dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^3 (6r^3 + 4r)\,dr\,d\theta$$

$$= \int_0^{2\pi} \left[\frac{6r^4}{4} + 2r^2\right]_0^3 d\theta$$

$$= \frac{279}{2}\int_0^{2\pi} d\theta = 279\pi$$



Example: Use the Divergence Theorem to find the outward flux of the vector field $\vec{F}(x,y,z) = x^3\hat{i} + y^3\hat{j} + z^3\hat{k}$ across the surface of the region that is enclosed by the hemisphere $z = \sqrt{a^2 - x^2 - y^2}$ and the plane z = 0

Solution: Let σ denote the outward-oriented surface and G the region that it encloses. The divergence of the vector field is $div\,\vec{F} = \frac{\partial}{\partial x}(x^3) + \frac{\partial}{\partial y}(y^3) + \frac{\partial}{\partial z}(z^3) = 3x^2 + 3y^2 + 3z^2$

Therefore, the flux across σ is

$$\iint_{\sigma} \vec{F}\cdot\hat{n}\,dS = \iiint_{G}\left(3x^2+3y^2+3z^2\right)dV$$

$$= \int_0^{2\pi}\int_0^{\frac{\pi}{2}}\int_0^{a}\left(3\rho^2\right)\rho^2\sin\phi\,d\rho\,d\phi\,d\theta$$

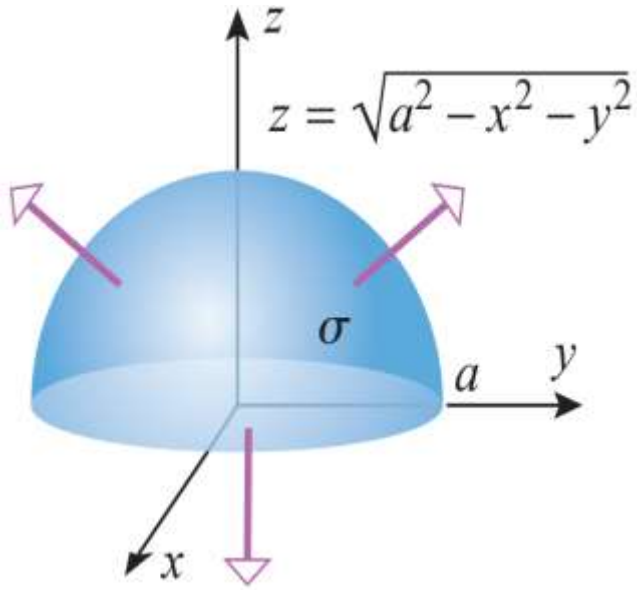
$$= 3\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\int_0^{a}\rho^4\sin\phi\,d\rho\,d\phi\,d\theta$$

$$= 3\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\left[\frac{\rho^5}{5}\right]_{\rho=0}^{a}\sin\phi\,d\phi\,d\theta$$

$$= \frac{3a^5}{5}\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\sin\phi\,d\phi\,d\theta$$

$$= \frac{3a^5}{5}\int_0^{2\pi}\left[-\cos\phi\right]_0^{\frac{\pi}{2}}d\theta$$

$$= \frac{3a^5}{5}\int_0^{2\pi}d\theta = \frac{6\pi a^5}{5}$$



Use the Divergence Theorem to find the flux of F across the surface σ with outward orientation.

1. $\vec{F}(x,y,z) = z^3\,\hat{i} - x^3\,\hat{j} + y^3\,\hat{k}$, where σ is the sphere $x^2+y^2+z^2=a^2$
2. $\vec{F}(x,y,z) = (x-z)\,\hat{i} + (y-x)\,\hat{j} + (z-y)\hat{k}$, where σ is the surface of the cylindrical solid bounded by $x^2+y^2=a^2$, $z=0$, and $z=1$.
3. $\vec{F}(x,y,z) = x\,\hat{i} + y\,\hat{j} + z\,\hat{k}$; σ is the surface of the solid bounded by the paraboloid $z=4-x^2-y^2$ and the xy-plane.
4. $\vec{F}(x,y,z) = x^3\,\hat{i} + y^3\,\hat{j} + z^3\,\hat{k}$; σ is the surface of the cylindrical solid bounded by $x^2+y^2=4$, $z=0$, and $z=3$.
5. $\vec{F}(x,y,z) = \left(x^3-e^y\right)\hat{i} + \left(y^3+\sin z\right)\hat{j} + \left(z^3-xy\right)\hat{k}$, where σ is the surface of the solid bounded above by $z=\sqrt{4-x^2-y^2}$ and below by the xy-plane
6. $\vec{F}(x,y,z) = 4xz\,\hat{i} + yz\,\hat{j} + z^2\,\hat{k}$, where σ is the surface of the solid bounded above by $z=\sqrt{a^2-x^2-y^2}$ and below by the xy-plane.

Stokes' Theorem: Let σ be a piecewise smooth oriented surface that is bounded by a simple, closed, piecewise smooth curve C with positive orientation.

If $\vec{F}(x,y,z) = f(x,y,z)\hat{i} + g(x,y,z)\hat{j} + h(x,y,z)\hat{k}$ where f, g, and h are continuous and have continuous first partial derivatives on some open set containing σ, and $\hat{n}$ is the unit normal vector, then $\oint_C \vec{F} \cdot d\vec{r} = \iint_\sigma \left(curl\ \vec{F}\right) \cdot \hat{n}\, dS$

Example: Verify Stokes' Theorem for the vector field $\vec{F}(x,y,z) = 2z\,\hat{i} + 3x\,\hat{j} + 5y\,\hat{k}$, taking σ to be the portion of the paraboloid $z = 4 - x^2 - y^2$ for which $z \geq 0$ with upward orientation, and C to be the positively oriented circle $x^2 + y^2 = 4$ that forms the boundary of σ in the xy-plane.

Solution: Since σ is oriented up, the positive orientation of C is counterclockwise looking down the positive z-axis. Thus, C can be represented parametrically (with positive orientation) by

$x = 2\cos t, \quad y = 2\sin t, \quad z = 0, \quad 0 \leq t \leq 2\pi$

Therefore,

$$\oint_C \vec{F} \cdot d\vec{r} = \oint_C 2z\,dx + 3x\,dy + 5y\,dz$$

$$= \int_0^{2\pi} \left[0 + (6\cos t)(2\cos t) + 0\right] dt$$

$$= \int_0^{2\pi} 12\cos^2 t\,dt$$

$$= 6\int_0^{2\pi} (1 + \cos 2t)\,dt = 6\left[t + \frac{\sin 2t}{2}\right]_0^{2\pi} = 12\pi$$

Again, $curl\ \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2z & 3x & 5y \end{vmatrix} = 5\hat{i} + 2\hat{j} + 3\hat{k}$

Since σ is oriented up and is expressed in the form $z = g(x, y) = 4 - x^2 - y^2$, it follows

$$\iint_{\sigma} \left(curl\ \vec{F}\right) \cdot \hat{n}\, dS$$

$$= \iint_{R} \left(curl\ \vec{F}\right) \cdot \left(-\frac{\partial z}{\partial x}\hat{i} - \frac{\partial z}{\partial x}\hat{j} + \hat{k} \right) dA$$

$$= \iint_{R} \left(5\hat{i} + 2\hat{j} + 3\hat{k}\right) \cdot \left(2x\hat{i} + 2y\ \hat{j} + \hat{k}\right) dA$$

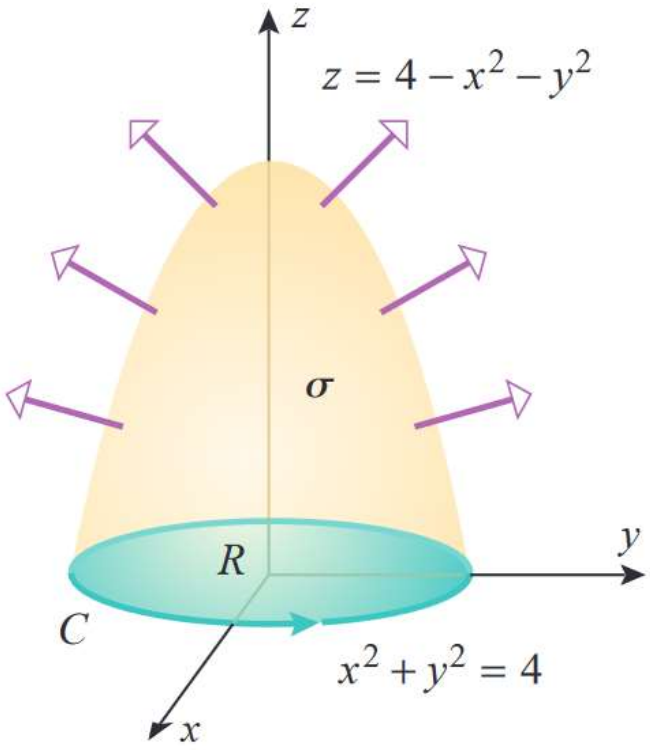
$$= \iint_{R} (10x + 4y + 3)\, dA$$

$$= \int_0^{2\pi} \int_0^2 (10r\cos\theta + 4r\sin\theta + 3)\, r\, dr\, d\theta$$

$$= \int_0^{2\pi} \left[10\frac{r^3}{3}\cos\theta + 4\frac{r^3}{3}\sin\theta + 3\frac{r^2}{2} \right]_0^2 d\theta$$

$$= \int_0^{2\pi} \left(\frac{80}{3}\cos\theta + \frac{32}{3}\sin\theta + 6 \right) d\theta$$

$$= \left[\frac{80}{3}\sin\theta - \frac{32}{3}\cos\theta + 6\theta \right]_0^{2\pi} = 12\pi$$



Verify Stokes' Theorem by evaluating the line integral and the surface integral. Assume that the surface has an upward orientation.

1. $\vec{F}(x, y, z) = 3z\,\hat{i} + 4x\ \hat{j} + 2y\hat{k}$, where σ is the portion of the paraboloid $z = 4 - x^2 - y^2$ above the xy-plane.

2. $\vec{F}(x, y, z) = (z - y)\,\hat{i} + (z + x)\ \hat{j} - (x + y)\hat{k}$; σ is the boundary of the paraboloid $z = 9 - x^2 - y^2$ above the xy-plane.